

$$f(a+h) - f(a) = Df(a)h + o(\|h\|)$$

Take $h = te_j$

$$\frac{f(a+te_j) - f(a)}{t} = \frac{Df(a)e_j + o(\|te_j\|)}{t}$$

$$\lim_{t \rightarrow 0} \frac{f(a+te_j) - f(a)}{t} = Df(a)e_j$$

$$\text{thus } \frac{\partial f}{\partial x_j} = Df(a)e_j$$

across all j columns
forms Jacobian

$$Df(a) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) \text{ gradient}$$

but if f is a vector
($f_1, \dots, f_m$)

$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

Since $D: \mathbb{R}^n \rightarrow \mathbb{R}^m$,
Jacobian + gradient
are linear maps
(linear approx)

Above h is vector by e_j

Take $h = tv$ w/ $\|v\|=1$

$$f(a+tv) - f(a) = tDf(a)v + o(\|tv\|)$$

$$\lim_{t \rightarrow 0} \frac{f(a+tv) - f(a)}{t} = Df(a)v$$

Directional Derivative

how to show existence of D ?
when is f differentiable?

Partial derivatives existing
does not imply f differentiable

E.g.: $f = \frac{xy}{x^2+y^2}$ if $x, y \neq 0$
otherwise $f=0$
if f not cont, not differentiable.
(by contin it is diff, then continuous)

But it partial der exists
& partials cont, f is diff?
then f differentiable

f is cont differentiable
 $\Rightarrow f$ differentiable &
all partials continuous

For $f: \mathbb{R}^n \rightarrow \mathbb{R}$
 $Df(a) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$ row vector

2nd order der (if exist)

$\frac{d}{dx_i} \left(\frac{\partial f}{\partial x_j} \right)$ Hessian
confirm subscript

Analysis

Total Derivative and Partial Derivatives

By definition of the Fréchet / total derivative for a function f :

$$f(a+h) - f(a) = Df(a)h + o(\|h\|)$$

Take $h = te_j$, where e_j is the j -th standard basis vector and $t \in \mathbb{R} \setminus \{0\}$:

$$\frac{f(a+te_j) - f(a)}{t} = \frac{Df(a)e_j + o(\|te_j\|)}{t}$$

Taking the limit as $t \rightarrow 0$:

$$\lim_{t \rightarrow 0} \frac{f(a+te_j) - f(a)}{t} = Df(a)e_j$$

Thus, the j -th partial derivative corresponds to the action of the derivative operator on the standard basis vector:

$$\frac{\partial f}{\partial x_j} = Df(a)e_j$$

Matrix Representation: Gradient and Jacobian

Across all j columns, this forms the Jacobian matrix.

- Scalar-valued function** ($f: \mathbb{R}^n \rightarrow \mathbb{R}$): The derivative is represented as the gradient (row vector):

$$Df(a) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

- Vector-valued function** ($f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, where $f = (f_1, \dots, f_m)$): The derivative becomes the $m \times n$ Jacobian matrix:

$$Df(a) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

Linear Approximation and Directional Derivatives

Since $Df(a) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ (a linear mapping from $\mathbb{R}^n \rightarrow \mathbb{R}^m$), the Jacobian and gradient represent the best local linear approximation of f at a .

Above, h was taken along the standard basis vectors (e_j). Now, generalize to an arbitrary unit direction vector.

Take $h = tv$ with $\|v\| = 1$ and $t \in \mathbb{R} \setminus \{0\}$:

$$f(a+tv) - f(a) = tDf(a)v + o(\|tv\|)$$

Dividing by t and taking the limit as $t \rightarrow 0$:

$$\lim_{t \rightarrow 0} \frac{f(a+tv) - f(a)}{t} = Df(a)v$$

The left-hand side is the **directional derivative** of f at a in the direction of v :

$$D_v f(a) = \nabla_v f(a) = Df(a)v$$

Differentiability and Continuous

Differentiability

How to show the existence of D ? When is f differentiable?

The existence of all partial derivatives does not imply that f is differentiable (or even continuous).

Counterexample:

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

- Along axes, the partial derivatives exist at $(0, 0)$: $\frac{\partial f}{\partial x}(0, 0) = 0$ and $\frac{\partial f}{\partial y}(0, 0) = 0$.
- However, f is not continuous at $(0, 0)$ (e.g., along the line $y = x$, $f(x, x) = \frac{1}{2} \neq 0$).
- Since f is not continuous, it cannot be differentiable (by the contrapositive of: differentiable $\Rightarrow$ continuous).

Sufficient Condition for Differentiability (C^1 Class)

If the partial derivatives of f exist in an open neighborhood of a point and are **continuous** at that point, then f is **differentiable** at that point.

(Answering the side note: it is the **partial derivatives** that must be continuous, which is a stronger condition than f merely being continuous).

- Continuously Differentiable (C^1):**

$f \in C^1$
 $\Leftrightarrow f$ is differentiable and all partial derivatives $\frac{\partial f}{\partial x_j}$ are continuous.

Second-Order Derivatives & The Hessian Matrix

For a scalar-valued function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the first derivative is given as the row vector (gradient):

$$Df(a) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

The Hessian Matrix ($D^2 f(a)$ or $H_f(a)$)

The second-order derivative (if it exists) is represented by the $n \times n$ Hessian matrix of mixed partial derivatives:

$$\frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

$$\left[\frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) \quad \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) \quad \dots \quad \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_1} \right) \right]$$

confirm subscripts

$$D^2 f(a) = \begin{bmatrix} \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) & \dots & \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_n} \right) \\ \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_2} \right) & \dots & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_n} \right) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_2} \right) & \dots & \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_n} \right) \end{bmatrix}$$

Hessian

$$D^2 f(a) = \begin{bmatrix} \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) & \dots & \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_n} \right) \\ \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_2} \right) & \dots & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_n} \right) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_2} \right) & \dots & \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_n} \right) \end{bmatrix}$$

then H symmetric

Hessian matrix of mixed partial derivatives:

$$\frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

$$D^2 f(a) = \begin{bmatrix} \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) & \dots & \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_n} \right) \\ \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_2} \right) & \dots & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_n} \right) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_2} \right) & \dots & \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_n} \right) \end{bmatrix}$$

Note on subscript convention (confirm subscripts):

- Element at row i , column j is $(H)_{ij} = \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) = \frac{\partial^2 f}{\partial x_j \partial x_i}$ (applying the Jacobian operator D to the row vector ∇f).
- In standard calculus notation, applying $\frac{\partial}{\partial x_i}$ first and then $\frac{\partial}{\partial x_j}$ is written as $\frac{\partial^2 f}{\partial x_j \partial x_i}$.

Symmetry of the Hessian (Clairaut's / Schwarz's Theorem)

$$\frac{\partial}{\partial x_j} \frac{\partial f}{\partial x_i} = \frac{\partial}{\partial x_i} \frac{\partial f}{\partial x_j}$$

If $f \in C^2$ (meaning all second-order partial derivatives exist and are continuous on an open set containing a), then mixed partial derivatives commute:

$$\frac{\partial^2 f}{\partial x_j \partial x_i} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

Consequently, the Hessian matrix H is **symmetric** ($H = H^T$).

For a scalar-valued function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the Hessian is always a square matrix of dimension $n \times n$.

- Domain Dimension:** The size of the matrix is determined strictly by the number of independent input variables n . Each row corresponds to differentiating with respect to x_i , and each column corresponds to differentiating with respect to x_j for $i, j \in \{1, \dots, n\}$.
- Second-Order Bilinear Form:** Structurally, the second Fréchet derivative $D^2 f(a)$ is a bilinear map on the tangent space: $D^2 f(a): \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$. Any matrix representation of a bilinear form on an n -dimensional vector space must be $n \times n$.

Extension to Vector-Valued Functions ($f: \mathbb{R}^n \rightarrow \mathbb{R}^m$): When generalizing to vector-valued mappings, the second derivative $D^2 f(a)$ is a multilinear map $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^m$, commonly represented as a 3-tensor of shape $m \times n \times n$. Each 2D "slice" corresponding to component function $f_k(k \in \{1, \dots, m\})$ remains an $n \times n$ square matrix:

$$H_{f_k} = \begin{bmatrix} \frac{\partial^2 f_k}{\partial x_1^2} & \dots & \frac{\partial^2 f_k}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f_k}{\partial x_n \partial x_1} & \dots & \frac{\partial^2 f_k}{\partial x_n^2} \end{bmatrix} \in \mathbb{R}^{n \times n}$$

Thus, the Hessian matrix is intrinsically square.

From <https://math.stackexchange.com/questions/1472046/hessian-matrix-is-always-square>.

Taylor series approx. to 2nd order

$$f(a+h) = f(a) + Df(a)h + \frac{1}{2}h^T D^2 f(a)h + o(\|h\|^2)$$

use 2nd der. to verify critical points in optimization

so when $Df(a) = 0$,

$$f(a+h) - f(a) = \frac{1}{2}h^T D^2 f(a)h + o(\|h\|^2)$$

Quadratic

so when $D^2 f(a)$ Hessian is symmetric, we can determine definiteness of H

2nd order conditions:

positive def $\Rightarrow$ local min

since $o(\|h\|^2) \rightarrow 0$

neg. def $\Rightarrow$ local max

but it's indefinite, no info

for ex: given f & point a w/ $h = \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix}$

$$f(x,y,z) = f(a) + Df(a)h$$

Second-Order Taylor Approximation & Optimization

Taylor Series Expansion to Second Order

For a twice continuously differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$:

$$f(a+h) = f(a) + Df(a)h + \frac{1}{2}h^T D^2 f(a)h + o(\|h\|^2)$$

Critical Points and Second-Order Conditions

Use the second derivative to classify critical points in optimization. At a critical point where $Df(a) = 0$:

$$f(a+h) - f(a) = \frac{1}{2}h^T D^2 f(a)h + o(\|h\|^2)$$

Since $D^2 f(a)$ (the Hessian matrix H) is symmetric, its definiteness dictates the local behavior of f :

- Positive Definite ($H \succ 0$):**

$$h^T H h > 0 \quad \forall h \neq 0 \Rightarrow \text{Strict Local Minimum}$$

(since the quadratic term dominates the higher-order error $o(\|h\|^2) \rightarrow 0$)

- Negative Definite ($H \prec 0$):**

$$h^T H h < 0 \quad \forall h \neq 0 \Rightarrow \text{Strict Local Maximum}$$

- Indefinite (H has both positive and negative eigenvalues):**

$\Rightarrow$ Saddle Point (no local extremum)

To calculate the remainder term $o(\|h\|^2)$, you substitute shifted function $f(a+h)$, or express it analytically using Taylor's theorem.

Methods to Calculate the Remainder

- Algebraic Remainder:** Compute $R_2(h) = f(a+h) - f(a) - Df(a)h - \frac{1}{2}D^2 f(a)h^2$.
- Lagrange Remainder Form:** Represent the error as $\frac{1}{6}D^3 f(\xi)h^3$ for some ξ on the line segment between a and $a+h$.

$$R_2(h) = \frac{1}{3!} \sum_{i,j,k=1}^n \frac{\partial^3 f}{\partial x_i \partial x_j \partial x_k} h_i h_j h_k \quad \text{for some } \xi \text{ on the line segment between } a \text{ and } a+h$$

Example Calculation

Given $f(x,y,z) = x^2 + y^2 + z^2$ and $a = (1,1,1)$, let $h = (h_1, h_2, h_3)$.

1. Evaluate $f(a)$

$$f(1,1,1) = 1^2 + 1^2 + 1^2 = 3$$

2. First Derivatives & Gradient $Df(a)$

$$\nabla f(x,y,z) = (2x, 2y, 2z) \Rightarrow \nabla f(1,1,1) = (2,2,2)$$

3. Second Derivatives & Hessian Matrix $D^2 f(a)$

$$D^2 f(a)h = 2h_1 + 2h_2 + 2h_3$$

tract the 2nd-order Taylor polynomial $P_2(h)$ from the exact
using third-order partial derivatives.

$+ h) - P_2(h)$, where $P_2(h) = f(a) + \nabla f(a)^T h + \frac{1}{2} h^T H f(a) h$.

exact error using third-order derivatives:

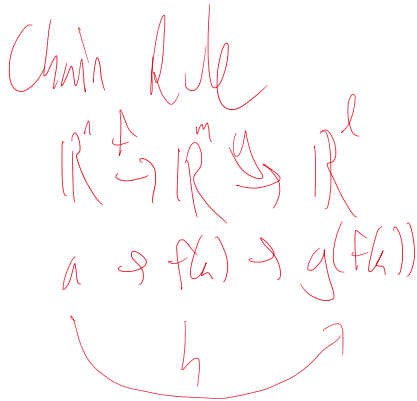
(0,1)

$$t = (h_1, h_2, h_3) = (x - 1, y - 1, z - 1).$$

$$f(x,y,z) = f(a) + Df(a)h + \frac{1}{2}h^T D^2 f(a)h + o(\|h\|^2)$$

$$\text{calculate } f(x,y,z) = e^{x^2+y^2+z^2}$$

how to
calc.
 $O(\|h\|^3)$?



$$h'(a) = g'(f(a)) \cdot f'(a)$$

$$\mathbb{R}^n \rightarrow \mathbb{R}^l$$

$$Dh(a)_{l \times n} = Dg[f(a)]_{l \times m} \cdot Df(a)_{m \times n}$$

$$\begin{bmatrix} f_{1,x_1} & f_{1,x_n} \\ \vdots & \vdots \\ f_{m,x_1} & f_{m,x_n} \end{bmatrix}$$

Inverse function thm

$f^{-1}(u)$ but when does
preimage of set inverse function exist?

$$f: U \rightarrow \mathbb{R}^n$$

mapping f must be

3. Second Derivative (Hessian):

$$\frac{\partial^2 f}{\partial x_i^2} = (2 + 4x_i^2)x_i^2, \quad \frac{\partial^2 f}{\partial x_i \partial x_j} = 4x_i x_j \delta_{ij}^2 \quad (i \neq j)$$

$$D^2 f(0,0,0) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = 2I_3$$

4. Taylor Expansion with $h = (x,y,z)^T$:

$$f(x,y,z) = 1 + \frac{1}{2}(x \ y \ z) \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + o(\|h\|^2)$$

$$f(x,y,z) = 1 + x^2 + y^2 + z^2 + o(x^2 + y^2 + z^2)$$

How to Calculate $o(\|h\|^2)$

The Little-o notation $o(\|h\|^2)$ is an asymptotic remainder term, not a single value to evaluate directly. To determine if bound it explicitly:

- Calculus (Asymptotic Vanishing):** By definition, any remainder $r_2(h)$ qualifies as $o(\|h\|^2)$ if

$$\lim_{h \rightarrow 0} \frac{r_2(h)}{\|h\|^2} = 0$$

- Lagrange Remainder (Explicit Form):** If $f \in C^3$, the remainder is given via the third-order derivative evaluated at some point $\xi = a + \zeta h$ for $\zeta \in (0,1)$:

$$R_2(h) = \frac{1}{6} \sum_{i,j,k=1}^3 \frac{\partial^3 f(\xi)}{\partial x_i \partial x_j \partial x_k} h_i h_j h_k$$

- Direct Series Expansion:** Using the 1D exponential series $e^u = 1 + u + \frac{u^2}{2!} + \dots$ with $u = \|h\|^2 = x^2 + y^2 + z^2$:

$$e^{\|h\|^2} = 1 + \|h\|^2 + \frac{\|h\|^4}{2} + \dots = 1 + \|h\|^2 + o(\|h\|^2)$$

Because $\frac{o(\|h\|^2)}{\|h\|^2} = o(\|h\|^2) \rightarrow 0$ as $h \rightarrow 0$, this confirms the error term is strictly $o(\|h\|^2)$.

From <https://www.geogebra.org/m/34729m437637259>

Multivariable Chain Rule

Setup and Composition of Mappings

Consider the composition $h = g \circ f$ defined across finite-dimensional Euclidean spaces:

$$\mathbb{R}^n \xrightarrow{f} \mathbb{R}^m \xrightarrow{g} \mathbb{R}^l$$

$$a \mapsto f(a) \mapsto g(f(a))$$

$$h: \mathbb{R}^n \rightarrow \mathbb{R}^l, \quad h(x) = (g \circ f)(x)$$

Total Derivative Formulation

Let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ be open sets such that $f(U) \subset V$. If f is differentiable at $a \in U$ and g is differentiable at $f(a) \in V$, then the composite mapping $h = g \circ f$ is differentiable at a , and its Fréchet derivative is given by the composition of linear maps:

$$h'(a) = g'(f(a)) \circ f'(a)$$

In terms of the derivative operator:

$$Dh(a) = Dg(f(a)) \cdot Df(a)$$

Matrix Representation & Dimensional Compatibility

Expressed via Jacobian matrices, the operation corresponds directly to standard matrix multiplication:

$$Dh(a) = Dg(f(a)) \cdot Df(a)$$

Where the inner matrix $Df(a)$ is the $m \times n$ Jacobian of f :

$$Df(a) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \begin{bmatrix} f_{1,x_1} & \dots & f_{1,x_n} \\ \vdots & \ddots & \vdots \\ f_{m,x_1} & \dots & f_{m,x_n} \end{bmatrix}$$

Component-wise / Coordinate Expansion

For the entry at index (i,k) of the resulting matrix $Dh(a)$, where $i \in \{1, \dots, l\}$ indexes the components of g and $k \in \{1, \dots, n\}$ indexes the independent variables of f :

$$\frac{\partial h_i}{\partial x_k}(a) = \sum_{j=1}^m \frac{\partial g_i}{\partial y_j}(f(a)) \frac{\partial f_j}{\partial x_k}(a)$$

This identity establishes that computing directional changes through composite systems amounts to computing standard linear transformations across successive tangent spaces $T_a \mathbb{R}^n \rightarrow T_{f(a)} \mathbb{R}^m \rightarrow T_{g(f(a))} \mathbb{R}^l$.

From <https://www.geogebra.org/m/34729m437637259>

$$f(x,y,z) = 1 + x^2 + y^2 + z^2$$

2. First Derivatives & Gradient $Df(a)$

$$\nabla f(x,y,z) = (2x, 2y, 2z) \implies \nabla f(1,1,1) = (2,2,2)$$

$$Df(a)h = 2h_1 + 2h_2 + 2h_3$$

3. Second Derivatives & Hessian Matrix $D^2 f(a)$

$$D^2 f(x,y,z) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\frac{1}{2} h^T D^2 f(a) h = \frac{1}{2} (2h_1^2 + 2h_2^2 + 2h_3^2) = h_1^2 + h_2^2 + h_3^2$$

4. Assemble the Taylor Polynomial $P_2(h)$

$$P_2(h) = 3 + 2(h_1 + h_2 + h_3) + (h_1^2 + h_2^2 + h_3^2)$$

5. Expand $f(a+h)$ directly

$$f(1+h_1, 1+h_2, 1+h_3) = (1+h_1)^2 + (1+h_2)^2 + (1+h_3)^2 +$$

$$= 3 + 2(h_1 + h_2 + h_3) + (h_1^2 + h_2^2 + h_3^2)$$

6. Compute the Remainder

$$R_2(h) = f(a+h) - P_2(h) = 0$$

Because $f(x,y,z)$ is a degree-2 polynomial, all third-order and higher terms vanish. The second-order Taylor expansion is exact, and the remainder is zero.

From <https://www.geogebra.org/m/34729m437637259>

Why the Chain Rule Reduces to 1D

1. The Derivative as a Best Local Linear Map

The derivative $Df(a)$ is defined explicitly as the linear map $L: \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfying:

$$\lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - L(h)\|}{\|h\|} = 0$$

Even when f and g are highly non-linear, their derivatives are linear maps between tangent spaces.

$$Df(a): \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad Dg(f(a)): \mathbb{R}^m \rightarrow \mathbb{R}^l$$

The composition of two linear maps is itself a linear map. The composition of two linear maps is itself a linear map.

$$D(g \circ f)(a) = Dg(f(a)) \cdot Df(a)$$

2. Arbitrary Chains of Composition (k-Layer)

For a composite chain of k functions:

$$F = f_k \circ f_{k-1} \circ \dots \circ f_1 \circ f_0$$

The total derivative matrix is strictly the ordered composition of the individual derivatives at each intermediate state:

$$DF(a_0) = J_{f_k}(a_{k-1}) \cdot J_{f_{k-1}}(a_{k-2}) \cdot \dots \cdot J_{f_1}(a_0) \cdot J_{f_0}(a_0)$$

What Happens for Highly Non-Linear Functions?

- Locality of the Matrices:** The Jacobian matrices are computed at the evaluation point a , and the relationship between infinitesimals da and df is strictly linear at that point.
- Higher-Order Effects (Curvature):** Non-linear functions have higher-order terms in their Taylor expansion ($D^2 f, D^3 f, \dots$). Computing $F(a)$ is a linear formula, which involves matrix products.

From <https://www.geogebra.org/m/34729m437637259>

$$(1+h_2)^2$$

l-order partial derivatives are zero ($\frac{\partial^l f}{\partial x_1 \partial x_1 \dots \partial x_l} = 0$). Therefore, the remainder $o(\|h\|^2) = 0$.

Matrix Multiplication

unique bounded linear operator $L \in$

derivatives at specific points and $f(u)$ are purely

$\mathbb{R}^n \rightarrow \mathbb{R}^m$ map. In coordinate space, the composition of matrix multiplication:

$$[Dg(f(x))][Df(x)]$$

Systems

product of k Jacobian matrices evaluated at their

se-mode automatic differentiation /

Linear Functions?

atrix is not constant across the domain, it Matrix multiplication computes the exact linearizat that specific point.

nearity appears in the higher-order terms of the higher derivatives of a composition requires bilinear tensor operations rather than simple

$f: U \rightarrow \mathbb{R}^n$ mapping f
 must be
 cont. diff.
 $\mathbb{R}^n$ with open
 same dimension

Take $a \in U$ & let Jacobian
 $n \times n$ $DF(a)$ be invertible & $f(a) = b$

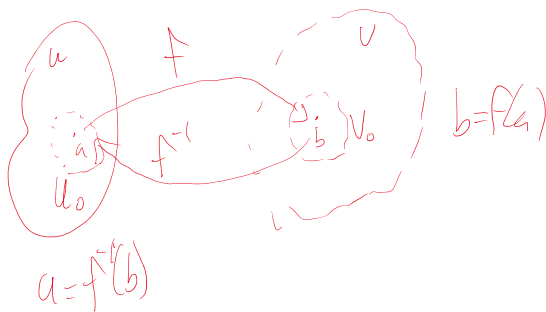
then $\exists U_0$ open set around a
 s.t. $a \in U_0$, $\exists V_0$ open set s.t. $b \in V_0$

$\Rightarrow f: U_0 \rightarrow V_0$ has inverse f^{-1}

f^{-1} cont. diff.

Jacobian $DF^{-1}(y)$ for $y \in V_0$ & $x \in U_0$

$$DF^{-1}(y) = [DF(x)]^{-1}$$



$$y = f(x)$$

$$f^{-1}(y) = f^{-1}(f(x)) = x$$

$$f^{-1}(f(x)) f'(x) = 1$$

$$f^{-1}(y) f'(x) = 1$$

Let $y = f(x)$, so that:

$$f^{-1}(y) = f^{-1}(f(x)) = x$$

Differentiating both sides with respect to x using the chain rule:

$$(f^{-1})'(f(x)) \cdot f'(x) = 1$$

$$(f^{-1})'(y) \cdot f'(x) = 1$$

If $f'(x) \neq 0$ at x , we can solve directly for the derivative of the inverse function:

$$(f^{-1})'(y) = \frac{1}{f'(x)}$$

(The derivative of the inverse equals 1 over the derivative).

From <https://www.khanacademy.org/multivariable-calculus/multivariable-differentiation/differentiating-vector-valued-functions/a/multivariable-chain-rule-derivative/a/multivariable-chain-rule-derivative>

Consider the strictly increasing function $f: (0, \infty) \rightarrow (0, \infty)$ defined by:

$$f(x) = x^3 + x$$

1. Direct Derivative of $f(x)$

$$f'(x) = 3x^2 + 1$$

Since $f'(x) \geq 1 > 0$ for all x , f is strictly monotonic and globally invertible on its domain.

2. Evaluate at a Specific Point

Choose $x = 2$. The corresponding output in the codomain is:

$$y = f(2) = 2^3 + 2 = 10 \Rightarrow f^{-1}(10) = 2$$

Evaluating the derivative of f at $x = 2$:

$$f'(2) = 3(2)^2 + 1 = 12 + 1 = 13$$

3. Apply the Inverse Derivative Formula

Using $(f^{-1})'(y) = \frac{1}{f'(x)}$ at $y = 10$ and $x = 2$:

$$(f^{-1})'(10) = \frac{1}{f'(2)} = \frac{1}{13}$$

4. Verification with an Explicitly Invertible Function

To see this match symbolic differentiation directly, consider $f(x) = x^3$ for $x > 0$:

- Direct derivative: $f'(x) = 3x^2$
- Explicit inverse: $f^{-1}(y) = y^{1/3}$
- Direct derivative of the inverse:

$$(f^{-1})'(y) = \frac{1}{3} y^{-2/3} = \frac{1}{3(y^{1/3})^2}$$

If $f'(x) \neq 0$ at x
 $f^{-1}(y) = \frac{1}{f'(x)}$ der. of inv equals 1 over der.
 Check example

$$(f^{-1})'(y) = \frac{1}{3}y^{-2/3} = \frac{1}{3(y^{1/3})^2}$$

• Substituting $y = x^3$:

$$(f^{-1})'(y) = \frac{1}{3(x)^2} = \frac{1}{f'(x)}$$

From <https://www.khanacademy.org/multivariable-calculus/multivariable-differentiation/a/multivariable-inverse-function-theorem/a/multivariable-inverse-function-theorem>

inverse is local statement around a point

check if neighb has to be open?

$$\mathbb{R}^n \supseteq U \xrightarrow{f} \mathbb{R}^n$$

ss

$Df(a)$ s.t. $f(a) = b$
 locally by linear approximation

locally must be invertible
 so inverse is well behaved

if $Df(a)$ invertible

Locality of the Inverse Function Theorem

The Inverse Function Theorem provides a strictly local statement around a point a , guaranteeing invertibility only on some neighborhood U_a of a , not necessarily globally across the entire domain U .

Does the Neighborhood Have to Be Open?

Yes, the neighborhoods U_a and V_a must be open sets.

• **Differentiability Requirement:** Differentiation is defined only on open sets (or at interior points) so that limits can be approached from every direction in $\mathbb{R}^n$.

• **Preservation of Openness:** Since f is a local diffeomorphism, it acts as an open mapping on U_a , ensuring that the image $V_a = f(U_a)$ contains an open ball around $b = f(a)$.

• **Well-Defined Inverse Derivative:** The inverse $f^{-1}: V_a \rightarrow U_a$ is differentiable on V_a , which formally requires its domain V_a to be open.

From <https://gemini.google.com/app/44750ec637bb770c>

Local Invertibility and Regularity

$$\mathbb{R}^n \supseteq U \xrightarrow{f} \mathbb{R}^n$$

Let $Df(a)$ be the derivative at a , with $f(a) = b$. Locally, by linear approximation if $Df(a)$ is invertible, then f must be locally invertible, ensuring the inverse mapping is well-behaved.

Inherited Regularity Properties ("Which?")

When $Df(a)$ is invertible:

• **Continuity:** The local inverse f^{-1} is continuous (homeomorphism).

• **Differentiability (C^k regularity):** If $f \in C^k$, then $f^{-1} \in C^k$ for any $k \geq 1$. If f is smooth (C^∞) or real analytic (C^ω), f^{-1} is likewise smooth or real analytic.

• **Open Mapping Property:** f maps open subsets within U_a to open subsets within V_a .

Counterexample: Invertibility Without Invertible Derivative

Consider $y = f(x) = x^3$:

• f is globally strictly increasing, differentiable, and invertible: $f^{-1}(y) = y^{1/3}$

• The derivative is $f'(x) = 3x^2 \Rightarrow f'(0) = 0$ (non-invertible linear map).

• The derivative of the inverse $(f^{-1})'(0)$ is undefined (vertical tangent), so f^{-1} fails to be differentiable at $y = 0$ despite f being differentiable and globally bijective.

Side Notes & Clarifications

• **Difference Between Derivative and Differentiation:**

◦ **Differentiation:** The operation/process of finding the rate of change or the linear mapping.

◦ **Derivative:** The resulting mathematical object (a number in $\mathbb{R}^1$, a linear map $Df(a) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, or its Jacobian matrix representation).

• **Connection with Fundamental Theorem of Calculus:**

◦ In 1D, the Fundamental Theorem connects the derivative to integration.

- **Difference between Derivative and Differentiation:**
 - **Differentiation:** The operation/process of finding the rate of change or the linear mapping.
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- **Connection with Fundamental Theorem of Calculus:**
 - In 1D, the Fundamental Theorem connects the derivative to integration via $\int_a^b f'(t) dt = f(b) - f(a)$.
 - In higher dimensions, this generalizes via **Stokes' Theorem** / the **Fundamental Theorem of Line Integrals**:

From <<https://gemini.google.com/app/44750ec637bb770c>>

Difference Between Derivative + Differentiation

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Implicit function thm

when output y is not clearly a function of input x

$F(x, y) = 0$

if $F: \mathbb{R}^2 \rightarrow \mathbb{R}$

when can we treat y as a function of x s.t. $y = y(x)$?

$$\text{then } F(x, y) = F(x, y(x)) = 0$$

$$F_1 + F_2 \cdot y'(x) = 0$$

$\frac{\partial}{\partial x}$ in x $\frac{\partial}{\partial y}$ in y

Partial derivatives $\nearrow$

$$y'(x) = -\frac{F_1}{F_2} \quad \text{if } F_2 \neq 0$$

Example?

$F: U \rightarrow \mathbb{R}^n$ Cont. diff

At

$\mathbb{R}^m \times \mathbb{R}^n$

for $x \in \mathbb{R}^m, y \in \mathbb{R}^n$

1. . . 1. $\mathbb{D}^n$

Implicit Function Theorem

Problem: Consider a relation defined by $F(x, y) = 0$, where $F: \mathbb{R}^2 \rightarrow \mathbb{R}$. When the variable y is not explicitly written as a function of x , when can we locally treat y as a well-defined function $y = y(x)$ such that:

$$F(x, y(x)) = 0$$

Derivation via Total Differentiation

Differentiating the identity $F(x, y(x)) = 0$ with respect to x using the multivariable chain rule:

$$F_1 + F_2 \cdot y'(x) = 0$$

Where:

- $F_1 = \frac{\partial F}{\partial x}$ is the partial derivative with respect to x .
- $F_2 = \frac{\partial F}{\partial y}$ is the partial derivative with respect to y .

Solving for $y'(x)$, provided $F_2 \neq 0$:

$$y'(x) = -\frac{F_1}{F_2} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}$$

Worked Example: Unit Circle

Consider the unit circle relation:

$$F(x, y) = x^2 + y^2 - 1 = 0$$

1. Compute Partial Derivatives:

$$F_1 = \frac{\partial F}{\partial x} = 2x, \quad F_2 = \frac{\partial F}{\partial y} = 2y$$

2. Check Condition for Local Function:

- If $F_2 = 2y \neq 0$ (i.e., $y \neq 0$), the Implicit Function Theorem guarantees the existence of a unique, differentiable function $y(x)$ locally.
- At the top hemisphere ($y > 0$), $y(x) = \sqrt{1 - x^2}$.
- At the bottom hemisphere ($y < 0$), $y(x) = -\sqrt{1 - x^2}$.

3. Compute the Derivative:

$$y'(x) = -\frac{F_1}{F_2} = -\frac{2x}{2y} = -\frac{x}{y}$$

4. Failure Points:

- At $(1, 0)$ and $(-1, 0)$, $F_2 = 0$. The tangent line is vertical, so y cannot be written locally as a single-valued function of x .

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we

ees

at be

$\mathbb{R}^{m \times n}$ for $x \in \mathbb{R}^m, y \in \mathbb{R}^n$

$$F(x, y) = 0 \in \mathbb{R}^n$$

Implicit function is

changing y to $y(x)$

so decomposing y in terms of x s.t. $y(x) \in \mathbb{R}^n$

x is input, y consequence or consequence

same dim as output

Implicit function is bad

so y is parameter?

treat x constant & calculate

if $D_y F(x_0, y_0)$ invertible

then $\exists U_0 \in \mathbb{R}^m$ w/ $x_0 \in U_0$ open

$V_0 \subseteq \mathbb{R}^n, y_0 \in V_0$ open

s.t. $\exists y = y(x) \quad U_0 \rightarrow V_0$

Multivariable Implicit Function Theorem General Framework

Let $U \subset \mathbb{R}^m \times \mathbb{R}^n$ be an open set, and

$F: U \rightarrow \mathbb{R}^n, (x, y) \mapsto F(x, y) = 0 \in \mathbb{R}^n$ where:

- $x \in \mathbb{R}^m$ acts as the vector of inputs
- $y \in \mathbb{R}^n$ acts as the vector of outputs
- The dimension of the output matches the dimension of the input

Theorem Statement

Let $(x_0, y_0) \in U$ satisfy $F(x_0, y_0) = 0$

If the partial Jacobian with respect to y is invertible at (x_0, y_0) :

- There exist open neighborhoods U_0 of x_0 and V_0 of y_0
- There exists a unique continuous function $y = y(x)$ defined on U_0 such that $y(x_0) = y_0$ and $F(x, y(x)) = 0$ for all $x \in U_0$

$$F(x, y(x)) = 0 \quad \forall x \in U_0$$

Derivative via the Chain Rule & Differentiating the Identity

Differentiating the identity $F(x, y(x)) = 0$ yields:

$$D_x F + D_y F \cdot D_x y = 0$$

$$\underbrace{D_x F}_{n \times m} + \underbrace{D_y F}_{n \times n} \cdot \underbrace{D_x y}_{m \times 1} = \underbrace{0}_{n \times 1}$$

Dimensional Check:

- $D_x F \in \mathbb{R}^{n \times m}$ (differentiating n outputs w.r.t. m inputs)
- $D_y F \in \mathbb{R}^{n \times n}$ (differentiating n outputs w.r.t. n inputs)
- $D_x y \in \mathbb{R}^{n \times m}$ (differentiating n outputs w.r.t. m inputs)
- Matrix product $(n \times n) \times (n \times m) = n \times m$

Solving for the Jacobian $D_x y(x)$:

$$D_x y(x) = -[D_y F(x, y(x))]^{-1} \cdot D_x F(x, y(x))$$

Worked Example: Two Equations

Consider the system with $x \in \mathbb{R}^1$ (input) and $y \in \mathbb{R}^2$ (output):

At the point $(x_0, y_0, z_0) = (1, 1, 1)$, both equations are satisfied.

1. Partial Jacobians:

$$D_{(y,z)} F = \begin{bmatrix} \frac{\partial F}{\partial y} & \frac{\partial F}{\partial z} \\ \frac{\partial G}{\partial y} & \frac{\partial G}{\partial z} \end{bmatrix} = \begin{bmatrix} 2y & 2z \\ 1 & 1 \end{bmatrix}$$

2. Evaluate at $(1, 1, 1)$:

$$D_{(y,z)} F(1, 1, 1) = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \Rightarrow \det \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} = 0$$

Here $D_{(y,z)} F$ is singular, so (y, z) cannot be uniquely determined as a function of x near $x = 1$.

3. Alternative Point Evaluation at $(1, 0, \sqrt{2})$:

$$D_{(y,z)} F = \begin{bmatrix} 2y & 2z \\ 1 & 1 \end{bmatrix} \Rightarrow D_{(y,z)} F(1, 0, \sqrt{2}) = \begin{bmatrix} 0 & 2\sqrt{2} \\ 1 & 1 \end{bmatrix}$$

$$\det(D_{(y,z)} F) = -2\sqrt{2} \neq 0$$

$$D_x F(1, 0, \sqrt{2}) = \begin{bmatrix} 2x \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

4. Compute Derivative of $\begin{bmatrix} y(x) \\ z(x) \end{bmatrix}$:

$$\begin{bmatrix} y'(1) \\ z'(1) \end{bmatrix} = -[D_{(y,z)} F(1, 0, \sqrt{2})]^{-1} \cdot D_x F(1, 0, \sqrt{2}) = \begin{bmatrix} -\sqrt{2} \\ 1 \end{bmatrix}$$

From <https://gemini.google.com/app>

theorem

and consider a continuously differentiable mapping:

$$\mathbb{R}^n$$

inputs / exogenous variables.

ogenous variables / outputs to be solved for.

ches the dimension of y (n equations in n unknowns).

o y , $D_y F(x_0, y_0) \in \mathbb{R}^{n \times n}$, is **invertible** ($\det D_y F(x_0, y_0) \neq 0$), then:

$U_0 \subset \mathbb{R}^m$ of x_0 and $V_0 \subset \mathbb{R}^n$ of y_0 such that $U_0 \times V_0 \subset U$.

ly differentiable mapping $y: U_0 \rightarrow V_0$ such that:

imensional Analysis

) = 0 with respect to x :

$$) = \mathbf{0}_{n \times m}$$

oordinate functions of F by m variables of x).

oordinate functions of F by n variables of y ; invertible).

omponents of y by m variables of x).

$n) = n \times m$, matching the dimension of $D_x F$ and the zero matrix.

$$x, y(x))$$

in Three Variables ($m = 1, n = 2$)

ut) and $(y, z) \in \mathbb{R}^2$ (endogenous outputs):

th equations vanish. Let $\mathbf{F} = \begin{bmatrix} F \\ G \end{bmatrix}$.

$$D_x \mathbf{F} = \begin{bmatrix} \frac{\partial F}{\partial x} \\ \frac{\partial G}{\partial x} \end{bmatrix} = \begin{bmatrix} 2x \\ 1 \end{bmatrix}$$

$$\begin{matrix} 2 \\ 1 \end{matrix} \Big] = 2(1) - 2(1) = 0$$

cannot be uniquely solved as functions of x near $(1, 1, 1)$.

$$(x_0, y_0, z_0) = (1, 0, \sqrt{2}) \text{ for } \mathbf{F} = \begin{bmatrix} x^2 + y^2 + z^2 - 3 \\ x + y - 1 \end{bmatrix}:$$

$$0, \sqrt{2}) = \begin{bmatrix} 0 & 2\sqrt{2} \\ 1 & 0 \end{bmatrix}$$

$$\begin{matrix} d\{\mathbf{bmatrix} \mathbf{x} \} = -\begin{matrix} \mathbf{bmatrix} \mathbf{0} & 2\sqrt{2} \end{matrix} \setminus \setminus 1 & 0 \\ \setminus \setminus 2 & \setminus \setminus 1 \end{matrix} \mathbf{bmatrix} \mathbf{x} = -\begin{matrix} \mathbf{bmatrix} \mathbf{0} & 1 \end{matrix} \setminus \setminus \frac{1}{\sqrt{2}} \setminus \setminus 2\sqrt{2} \} & \\ \setminus \setminus 1 \end{matrix} \mathbf{bmatrix} \mathbf{x} = \begin{matrix} \mathbf{bmatrix} \mathbf{-1} \end{matrix} \setminus \setminus -\frac{1}{\sqrt{2}} \setminus \setminus \sqrt{2} \}$$

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$$- \quad \nabla y = y(x) \quad \text{as } - \quad \text{as } - \quad \text{as } -$$

& $F(x, y(x)) = 0$ holds locally
 $y(x)$ must be cont diffiable

apply chain rule

$$\begin{array}{c} D_x F + D_y F D_x y \\ \begin{array}{c} n \times m \\ \times \quad \bar{F}_1 \dots \bar{F}_m \end{array} \quad \begin{array}{c} n \times n \\ \uparrow \\ \text{for } x \end{array} \quad \begin{array}{c} n \times m \\ \uparrow \\ \text{for } y \end{array} \end{array}$$

↑ Check dimensions & why

$$D_x F + D_y F D_x y = 0$$

$$\Rightarrow D_x y(x) = -[D_y F]^{-1} [D_x F]$$

ex. given

$$F(x, y, z) = 0 \in \mathbb{R}$$

$$G(x, y, z) = 0 \in \mathbb{R}$$

$$\Rightarrow \begin{cases} y(x) \\ z(x) \end{cases}$$

if x input
 & y, z param

Example?

—

$f(y, z)$ param

involve implicit funct if
cannot solve for y explicitly

Lecture Summary: Multivariable Differential Calculus

- **Total Derivative & Linear Approximation:** The Fréchet derivative $Df(a)$ is the unique bounded linear operator approximating the local behavior of f . Its matrix representation corresponds to the gradient ($\mathbb{R}^n \rightarrow \mathbb{R}$) or the Jacobian ($\mathbb{R}^n \rightarrow \mathbb{R}^m$).
- **Directional Derivatives:** Directional rates of change along any unit vector v are obtained via matrix-vector multiplication: $D_v f(a) = Df(a)v$.
- **Differentiability vs. Partial Existence:** The existence of partial derivatives does not guarantee differentiability or continuity. Continuous partial derivatives (C^1 regularity) provide the standard sufficient condition for differentiability.
- **Second-Order Structure & Optimization:** For $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the second derivative is represented by the symmetric $n \times n$ Hessian matrix $H_f(a)$ when $f \in C^2$ (Clairaut's Theorem). Local extrema at critical points ($Df(a) = 0$) are classified by the definiteness of $H_f(a)$: positive definite yields a strict local minimum, negative definite yields a strict local maximum, and indefinite yields a saddle point.
- **Multivariable Chain Rule:** Differentiation of composite mappings corresponds to the sequential multiplication of their Jacobian matrices: $D(g \circ f)(a) = Dg(f(a)) \cdot Df(a)$.
- **Inverse Function Theorem:** If $f: U \rightarrow \mathbb{R}^n$ is C^1 and $Df(a)$ is invertible ($\det Df(a) \neq 0$), f is a local diffeomorphism with $D(f^{-1})(y) = [Df(x)]^{-1}$.
- **Implicit Function Theorem:** Given an equilibrium relation $F(x, y) = 0$ with $F: \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, invertibility of the partial Jacobian $D_y F$ allows locally expressing y as a differentiable function of x , with derivative $D_x y = -[D_y F]^{-1} D_x F$.

Core Takeaways for Graduate Study

- **Always Track Domain & Codomain Dimensions:** When setting up Jacobians, chain rules, or implicit derivatives, verify dimensional consistency before calculating. An $n \times m$ derivative matrix operating on an $m \times 1$ perturbation vector must map conformably into $\mathbb{R}^n$.
- **Differentiability Is a Global/Neighborhood Property:** Do not assume differentiability solely from directional or partial derivatives; verify that the partials are continuous (C^1) on an open neighborhood.
- **The Hessian Is Inherently a Bilinear Form:** Because $D^2 f(a)$ maps pairs of tangent vectors into $\mathbb{R}$, its matrix representation for a scalar field is always square ($n \times n$).
- **Standardize Invertibility Conditions:** The non-singularity condition $\det D \neq 0$ is the unifying thread connecting local invertibility (Inverse Function Theorem) and the solvability of endogenous systems in comparative statics (Implicit Function Theorem).

From <<https://gemini.google.com/app/44750ec637bb770c>>

